

3D Trajectory Optimization for UAV-assisted Covert Visible Light Communication

Tamunoene Eugene Bamson, Liang Zhang, *Member, IEEE*

Department of Engineering, University of Maryland Eastern Shore, Princess Anne, MD, 21853, USA

Email: {tbamson, lzhang1}@umes.edu

Abstract—Unmanned aerial vehicle (UAV)-assisted visible light communication (VLC) has emerged as a promising paradigm for high-data-rate and secure wireless communications due to its abundant spectrum and inherent line-of-sight (LoS) characteristics. In this paper, we investigate a covert UAV-enabled VLC system in which a UAV transmitter communicates with a legitimate ground user while remaining undetectable to a passive warden. To ensure covertness, a Kullback–Leibler (KL) divergence-based constraint is adopted to limit the warden’s detection capability. We formulate a joint optimization problem to maximize the average covert throughput by optimizing the UAV’s three-dimensional (3D) trajectory and transmission intensity under trajectory, optical power transmission intensity, and covertness constraints. The studied problem is non-convex due to the coupled trajectory and communication variables. To address this challenge, we develop an efficient iterative framework based on successive convex approximation (SCA) and block coordinate descent (BCD), which decomposes the original problem into tractable subproblems. Numerical results demonstrate that the proposed scheme significantly improves covert throughput compared to baseline strategies, while strictly satisfying the covertness requirement.

I. INTRODUCTION

Unmanned aerial vehicles (UAVs) have emerged as agile platforms for on-demand wireless connectivity due to their high mobility, rapid deployment, and flexible coverage [1]. These features make UAVs particularly attractive for emergency response, temporary events, and dense urban environments where conventional infrastructure is limited [2]. However, the increasing demand for high data rates and spectrum efficiency exposes the limitations of traditional radio-frequency communications and UAV communications, including spectrum scarcity and electromagnetic interference.

To address these challenges, visible light communication (VLC) has recently gained attention as a promising complementary technology. By leveraging the vast unlicensed optical spectrum, VLC enables high-rate transmissions without RF interference while supporting low-cost implementations using LEDs and photodetectors [3]. The integration of UAVs with VLC further enhances system performance by combining UAV mobility with the line-of-sight (LoS)-dominant optical channel, enabling high-capacity and interference-free air-to-ground links. As a result, UAV-assisted VLC systems are well suited

for next-generation wireless networks, offering both communication and illumination capabilities.

Recent works have validated the feasibility and performance potential of UAV-enabled VLC. Experimental testbeds have demonstrated reliable air-to-ground optical links, confirming practical deployment viability [3]. On the theoretical side, significant efforts have focused on optimizing UAV deployment and resource allocation to exploit LoS channel characteristics and improve throughput, energy efficiency, and quality of service. For instance, deep learning-based deployment strategies have been proposed to optimize UAV placement under communication and illumination constraints [4], while NOMA-enabled designs have been investigated to enhance system capacity and spectral efficiency in UAV-assisted VLC networks [5]. Despite these advances, most studies primarily target performance metrics such as coverage and sum-rate.

Beyond performance optimization, security has become a critical concern in UAV-assisted VLC systems. Due to the broadcast and LoS-dominant nature of optical channels, transmitted signals are inherently exposed to potential eavesdropping. Physical-layer security (PLS) techniques have therefore been explored to improve secrecy performance through trajectory design, user association, and cooperative strategies [6]. However, secrecy alone is insufficient in scenarios requiring a higher level of privacy. In contrast, covert communication aims to hide the very existence of transmission, making it fundamentally more stringent and challenging.

Covert communication in VLC introduces unique challenges compared to RF systems. Optical signals must satisfy non-negativity, peak, and average intensity constraints, which fundamentally alter system design and achievable performance limits [7]. Recent studies have begun to address these challenges by analyzing covert VLC performance, optimizing UAV altitude, and incorporating advanced techniques such as reconfigurable intelligent surfaces (RIS) and cooperative jamming [8]–[10]. Moreover, UAV mobility has been recognized as a key enabler for improving covert performance by dynamically shaping channel conditions and detection probabilities.

Despite these advancements, several important gaps remain. First, many existing works restrict UAV motion to fixed altitude or two-dimensional trajectories, limiting the exploitation of full three-dimensional (3D) mobility. Second, recent 3D covert

Corresponding author: Liang Zhang.

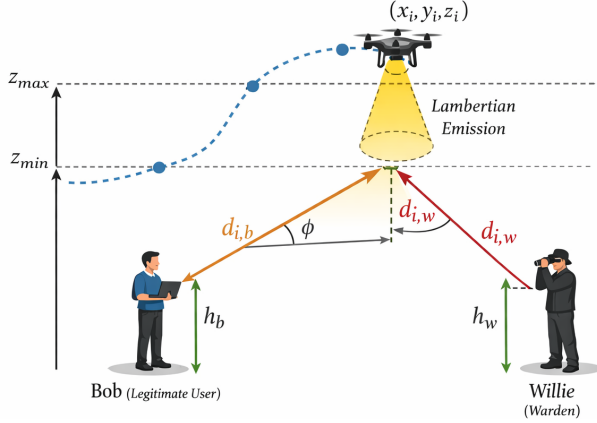


Fig. 1. UAV-based covert VLC system model.

VLC frameworks often rely on multi-UAV cooperation, jamming, or complex multiple access schemes, which obscure the fundamental tradeoff between UAV trajectory, optical intensity, and covertness. Third, although fundamental covert limits for optical systems have been established, their integration into practical UAV trajectory design remains largely unexplored.

In this paper, we jointly optimize the UAV's horizontal trajectory, altitude, and time-varying optical transmission intensity to maximize long-term covert throughput under a rigorous Kullback–Leibler (KL) divergence constraint for a single-UAV-assisted covert VLC system. By explicitly incorporating optical intensity constraints into the covertness formulation, this work bridges the gap between fundamental covert optical communication theory and practical UAV-enabled VLC system design.

II. SYSTEM MODEL

We consider a single-UAV visible light communication (VLC) system in the presence of a passive warden. As illustrated in Fig. 1, a UAV-mounted light-emitting diode (LED) transmitter (Alice) communicates with a legitimate ground user (Bob), while a warden (Willie) attempts to detect the existence of transmission. This setup follows standard UAV–VLC architectures and covert communication models [3] [8].

A. Network Topology and Geometry

The UAV flies over a finite horizon T , discretized into I time slots of duration $\Delta t = \frac{T}{I}$. The UAV position in slot $i \in \{1, \dots, I\}$ is

$$\mathbf{q}_i = (x_i, y_i, z_i), \quad (1)$$

while Bob and warden w are located at fixed ground coordinates $\mathbf{u}_b = (x_b, y_b, 0)$ and $\mathbf{u}_w = (x_w, y_w, 0)$, respectively. UAV kinematics impose

$$\|\mathbf{q}_{i+1} - \mathbf{q}_i\|_2 \leq V_{\max} \Delta t, \quad z_{\min} \leq z_i \leq z_{\max}, \quad (2)$$

which are widely adopted in UAV trajectory optimization frameworks [4]–[6].

B. Lambertian VLC Channel Model

We adopt a line-of-sight (LoS) Lambertian channel model, which is appropriate for air-to-ground VLC links due to dominant LoS propagation [4], [5]. Let $d_{i,k}$ denote the distance between the UAV and node $k \in \{b, w\}$ in slot i . Assuming perpendicular LED/PD alignment, the channel gain is

$$h_{i,k} = h_0 \frac{z_i^{m+1}}{d_{i,k}^{m+3}}, \quad (3)$$

where m is the Lambertian order and h_0 collects optical device constants.

C. Optical Signaling Model and Intensity Constraints

The UAV employs intensity modulation/direct detection (IM/DD). The transmitted optical signal X_i satisfies

$$0 \leq X_i \leq A_{\max}, \quad \mathbb{E}[X_i] \leq \bar{A}. \quad (4)$$

The received signal at node k is

$$Y_{i,k} = r_k h_{i,k} X_i + N_{i,k}, \quad (5)$$

where $N_{i,k} \sim \mathcal{N}(0, \sigma_k^2)$.

D. Achievable Throughput at the Legitimate User

Using entropy-based bounds for IM/DD channels, the mutual information per slot admits the lower bound

$$I(X_i; Y_{i,b}) \geq \frac{1}{2} \log_2 \left(1 + \frac{r_b^2 h_{i,b}^2 \mathbb{E}[X_i^2]}{2\pi e \sigma_b^2} \right), \quad (6)$$

The per-slot throughput is

$$R_i = B I(X_i; Y_{i,b}), \quad (7)$$

B is the optical bandwidth, and r_b^2 is the photodetector responsivity and the long-term average throughput is

$$\bar{R} = \frac{1}{I} \sum_{i=1}^I R_i. \quad (8)$$

E. Covertness Formulation via KL Divergence

Willie performs binary hypothesis testing:

$$\mathcal{H}_0 : Y_{i,w} = N_{i,w}, \quad \mathcal{H}_1 : Y_{i,w} = r_w h_{i,w} X_i + N_{i,w}. \quad (9)$$

Covertness is enforced using the KL-divergence constraint

$$D_i = N D(p_1^{(i)} \| p_0^{(i)}), \quad (10)$$

with total budget

$$\sum_{i=1}^I D_i \leq 2\epsilon^2. \quad (11)$$

This relative-entropy constraint is the same fundamental form used to connect covert feasibility to detection limits, and it is especially natural for optical channels because it integrates non-negativity and peak/average intensity constraints into the statistical distinguishability analysis [7]. It is also consistent with

KL-based covert VLC analysis used for UAV height optimization [8]. While some recent UAV-VLC covert designs incorporate jamming and analyze detection-error probability explicitly [10], the KL budget employed here directly captures a rigorous information-theoretic covertness requirement without relying on cooperative jamming.

III. PROBLEM FORMULATION

The overall problem is formulated as

$$\max_{\mathbf{Q}, \mathbf{p}} \quad \frac{1}{I} \sum_{i=1}^I \frac{B}{2} \log_2 \left(1 + \kappa h_{i,b}^2 p_i \right) \quad (12)$$

$$\text{s.t.} \quad \|\mathbf{q}_{i+1} - \mathbf{q}_i\|_2 \leq V_{\max} \Delta t, \quad \forall i, \quad (13)$$

$$z_{\min} \leq z_i \leq z_{\max}, \quad \forall i, \quad (14)$$

$$\mathbf{q}_1 = \mathbf{q}_{\text{init}}, \quad \mathbf{q}_I = \mathbf{q}_{\text{final}}, \quad (15)$$

$$0 \leq p_i \leq p_{\max}, \quad \forall i, \quad (16)$$

$$\frac{1}{I} \sum_{i=1}^I p_i \leq \bar{p}, \quad (17)$$

$$N\alpha \sum_{i=1}^I h_{i,w}^2 p_i \leq 2\epsilon^2. \quad (18)$$

The problem is highly non-convex due to:

- 1) The nonlinear dependence of $h_{i,b}$ and $h_{i,w}$ on q_i
- 2) The multiplicative coupling between channel gains and intensity variables.
- 3) The KL-based constraint that also depends on squared channel gains.

Similar non-convexity arises in UAV trajectory optimization [5] [10], but here it is compounded by the information-theoretic covertness constraint derived from optical covert communication theory [7]. This makes the problem fundamentally different from secrecy-rate or QoS-driven formulations.

IV. TRAJECTORY OPTIMIZATION ALGORITHM

Based on the system model in Section II, we formulate a joint 3D trajectory and optical intensity optimization problem for UAV-assisted covert VLC. The objective is to maximize the long-term average covert throughput subject to UAV mobility, optical intensity, and covertness constraints.

A. Optimization Variables

The optimization variables are:

- The UAV trajectory:

$$\mathbf{Q} \triangleq \{\mathbf{q}_i\}_{i=1}^I \quad (19)$$

- Optical intensity (second moment):

$$\mathbf{p} \triangleq \{p_i\}_{i=1}^I \quad (20)$$

Unlike prior UAV-VLC works that restrict motion to fixed altitude or 2D trajectories [4], [5], we consider full 3D mobility.

Moreover, in contrast to multi-user or jammer-assisted designs [9] [10], a single-user setting is adopted to isolate the fundamental trajectory-covert tradeoff.

B. Objective Function: Long-Term Covert Throughput

Using the lower bound in (6), the achievable rate per slot is:

$$R_i = \frac{B}{2} \log_2 \left(1 + \kappa h_{i,b}^2 p_i \right), \quad (21)$$

Here, $\kappa = \frac{r_b^2}{2\pi e \sigma_b^2}$. $h_{i,b}$ follows the Lambertian channel model widely used in [4]–[6].

The long-term average covert throughput is

$$\bar{R}(\mathbf{Q}, \mathbf{p}) = \frac{1}{I} \sum_{i=1}^I \frac{B}{2} \log_2 \left(1 + \kappa h_{i,b}^2 p_i \right). \quad (22)$$

This objective resembles sum-rate maximization problems in the UAV-VLC system [4], [5], but differs fundamentally in that it is constrained by statistical detectability rather than SINR or QoS metrics.

C. Constraints

- 1) UAV Mobility Constraints: The UAV must satisfy (2). Which are standard in UAV trajectory optimization [4]–[6].
- 2) Optical Intensity Constraints: Due to IM/DD signaling and eye-safety requirements inherent to VLC systems, (4). Which reflect non-negativity and peak/average constraints unique to VLC systems [7] [8].
- 3) KL-Divergence-Based Covertness Constraint To ensure covert transmission, the KL-divergence constraint in (11) is enforced. Using standard upper bounds for IM/DD channels [6], this yields the surrogate

$$N\alpha \sum_{i=1}^I h_{i,w}^2 p_i \leq 2\epsilon^2 \quad (23)$$

This constraint ensures that the warden cannot reliably detect the presence of transmission, in contrast to secrecy-based designs that focus on decoding capability [6]. Unlike cooperative jamming approaches [10], covertness is achieved here solely through trajectory and power control.

V. SCA-BASED OPTIMIZATION FRAMEWORK

The problem in Section III is non-convex due to the nonlinear dependence of the Lambertian channel gains on the UAV trajectory and their multiplicative coupling with the intensity variables. Similar challenges arise in UAV trajectory optimization and joint resource allocation problems [4]–[6]. However, the inclusion of a KL-divergence-based covertness constraint fundamentally distinguishes this formulation from secrecy-rate or QoS-driven designs.

To address this, we adopt a block coordinate descent (BCD) framework combined with successive convex approximation (SCA), which has been widely used in UAV-enabled communication systems [5], [6], [10].

Algorithm 1: SCA-Based 3D Covert Trajectory and Intensity Optimization

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- 1: Initialize feasible $Q^{(0)}, A^{(0)}$
 - 2: Set $t \leftarrow 0$
 - 3: **repeat**
 - 4: Construct concave surrogate $\tilde{R}_i(q_i)$
 - 5: Construct convex upper bound $\tilde{D}_i(q_i)$
 - 6: Solve convex trajectory subproblem to obtain $Q^{(t+1)}$
 - 7: Construct concave surrogate $\tilde{R}_i(A_i^2)$
 - 8: Construct convex divergence upper bound $\tilde{D}_i(A_i^2)$
 - 9: Solve convex intensity subproblem to obtain $A^{(t+1)}$
 - 10: $t \leftarrow t + 1$
 - 11: **until** Convergence criterion satisfied
 - 12: Output Q^*, A^*
-

A. Block Coordinate Descent Decomposition

Let the optimization variables be the UAV trajectory

$$\mathbf{Q} = \{\mathbf{q}_i\}_{i=1}^I \quad (24)$$

and the intensity vector

$$\mathbf{p} = \{p_i\}_{i=1}^I. \quad (25)$$

We decompose the original problem into two subproblems:

- 1) **Intensity optimization** for fixed trajectory $\mathbf{Q}$;
- 2) **Trajectory optimization** for fixed intensity $\mathbf{p}$.

This alternating structure mirrors the joint trajectory–power optimization adopted in [5], [10], in which the trajectory and communication variables are updated sequentially to address the strong coupling.

Unlike previous works that optimize the sum-rate or minimum user rate, our formulation embeds a KL-divergence-based constraint derived from [7]–[9], thereby introducing a statistical detectability condition rather than a secrecy-SINR constraint, as in [6].

B. Intensity Optimization Subproblem

For a fixed trajectory $\mathbf{Q}$, the channel gains $h_{i,b}$ and $h_{i,w}$ become constants. The intensity subproblem reduces to

$$\max_{\mathbf{p}} \sum_{i=1}^I \log \left(1 + \kappa h_{i,b}^2 p_i \right) \quad (26)$$

$$(27)$$

subject to linear constraints (14)–(16). This is a convex problem since the objective is concave in p_i and all constraints are linear. It can therefore be solved optimally using standard convex optimization methods. Similar convex power allocation problems appear in UAV–VLC systems [4], [5], the KL-based constraint directly reflects covert communication limits [7] [8].

C. Trajectory Optimization via SCA

For fixed intensity $\mathbf{p}$, the trajectory optimization remains non-convex because the squared channel gains

$$g_{i,k}(\mathbf{q}_i) = h_{i,k}^2 \quad (28)$$

are nonlinear functions of $\mathbf{q}_i$ under the Lambertian LoS model used in [4]–[6].

To address this, we apply the Taylor first-order approximation to linearize $g_{i,k}(\mathbf{q}_i)$ around the current iterate $\mathbf{q}_i^{(t)}$:

$$g_{i,k}(\mathbf{q}_i) \approx g_{i,k}(\mathbf{q}_i^{(t)}) + \nabla g_{i,k}(\mathbf{q}_i^{(t)})^\top (\mathbf{q}_i - \mathbf{q}_i^{(t)}). \quad (29)$$

This first-order SCA linearization is widely adopted in UAV-enabled wireless optimization [5], [6], [10] and in RIS-assisted covert VLC systems where nonlinear channel terms are convexified iteratively [9].

Defining the affine approximations by $\tilde{g}_{i,b}$ and $\tilde{g}_{i,w}$, the trajectory subproblem becomes

$$\max_{\mathbf{Q}} \sum_{i=1}^I \log \left(1 + \kappa \tilde{g}_{i,b} p_i \right) \quad (30)$$

$$\text{s.t. } N\alpha \sum_{i=1}^I \tilde{g}_{i,w} p_i \leq 2\epsilon^2, \quad (31)$$

$$\text{UAV mobility constraints.} \quad (32)$$

Because the logarithm is concave and $\tilde{g}_{i,b}$ is affine in $\mathbf{q}_i$, the objective becomes concave. The KL surrogate constraint is linear, and the mobility constraints form second-order cones. Therefore, each SCA step solves a convex program.

Importantly, while [8] considers optimal UAV height for covert VLC under KL constraints, our formulation extends this to full 3D trajectory optimization, significantly enlarging the feasible design space.

D. Trust-Region Stabilization

To ensure numerical stability, we impose a trust-region constraint:

$$\|\mathbf{q}_i - \mathbf{q}_i^{(t)}\|_2 \leq \Delta. \quad (33)$$

Trust-region SCA has been employed in joint trajectory–resource optimization problems for [5] [9] [10] to prevent divergence caused by aggressive linearization.

This constraint guaranties that the linear approximation remains locally accurate and ensures a stable, monotonic improvement in the objective.

E. Convergence Analysis

At each BCD iteration:

- 1) The intensity subproblem is solved optimally (convex).
- 2) The trajectory subproblem maximizes a tight concave surrogate that matches the original objective in the current iteration.

Since:

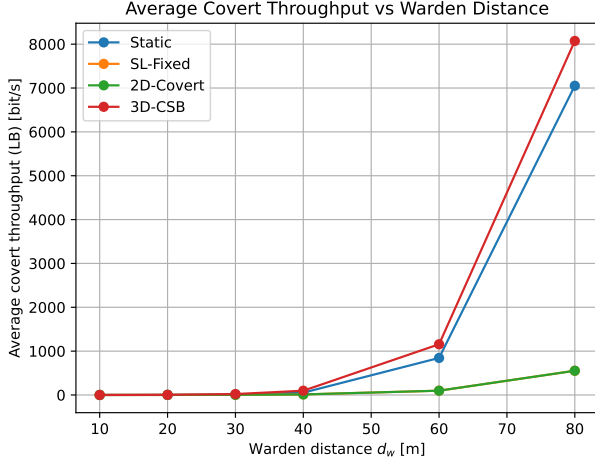


Fig. 2. Average covert throughput $\bar{R}$ versus horizontal warden distance d_w .

- The surrogate function is a global lower bound,
- The objective is non-decreasing across iterations,
- The feasible set is compact due to mobility and intensity constraints,

the proposed BCD–SCA algorithm converges to a stationary point of the original non-convex problem.

Similar convergence arguments are used in [5] [9] [10].

VI. NUMERICAL RESULTS

This section evaluates the performance of the proposed 3D-covert trajectory optimization algorithm and compares it with three representative baselines:

- 1) **Static Hover:** UAV hovers above Bob with optimized altitude and intensity under the same KL constraint.
- 2) **SL-Fixed:** UAV follows a straight-line trajectory at fixed altitude; only intensity is optimized.
- 3) **2D-Covert:** UAV optimizes horizontal coordinates and intensity while altitude remains fixed.
- 4) **Proposed 3D-Covert:** UAV jointly optimizes horizontal position, altitude, and intensity.

Unless otherwise stated, the system parameters follow those in the system model section. The flight horizon is $T = 100$ s with $I = 100$ slots, the Lambertian order is $m = 1$, and the covertness constraint is enforced via the KL surrogate

$$\sum_{i=1}^I N a h_{i,w}^2 p_i \leq 2\epsilon^2. \quad (34)$$

A. Average Covert Throughput vs Warden Distance

Fig. 2 shows the average covert throughput $\bar{R}$ versus the horizontal distance between Bob and Willie.

First, throughput increases with warden distance. As Willie moves farther away, the channel gain h_w decreases due to Lambertian path loss, which relaxes the KL-divergence constraint and allows higher transmission intensity.

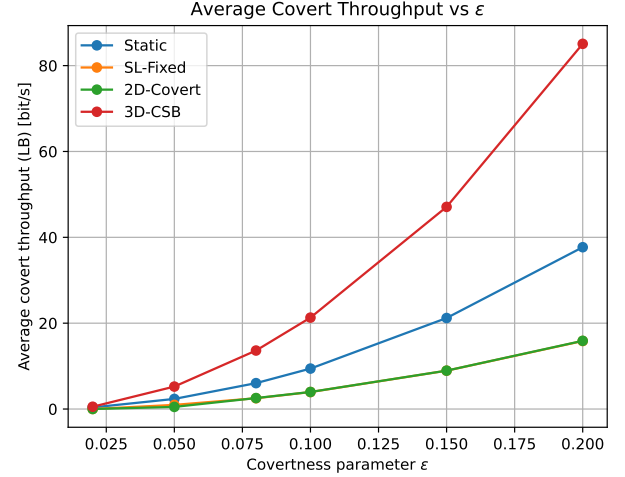


Fig. 3. Average covert throughput $\bar{R}$ versus covertness parameter ϵ .

Second, the proposed 3D-Covert consistently outperforms all baselines. When the warden is close, all schemes are tightly constrained, and gains are moderate. However, at intermediate distances, 3D-Covert achieves significant improvements (approximately 20–35%) over SL-Fixed and 2D-Covert by jointly optimizing horizontal and vertical geometry.

Finally, as the warden distance becomes large, the covertness constraint becomes inactive and all schemes converge toward similar throughput levels, though 3D-Covert still maintains a slight advantage due to improved geometric alignment.

B. Average Covert Throughput vs Covertness Parameter ϵ

Fig. 3 illustrates $\bar{R}$ as a function of the covertness parameter ϵ .

Throughput increases monotonically with ϵ , since a larger divergence budget permits higher transmission intensity. In the stringent covert regime (small ϵ), all schemes are power-limited and performance differences are small.

In the moderate regime, the advantage of trajectory optimization becomes pronounced, with 3D-Covert achieving up to 30–40% improvement over Static Hover and about 25–30% over SL-Fixed. As ϵ increases further, all schemes approach the non-covert limit, consistent with fundamental covert communication behavior.

C. Effect of Altitude Freedom

To isolate the impact of vertical mobility, we compare 3D-Covert with 2D-Covert under varying altitude ranges.

When the allowable altitude range is limited, both schemes perform similarly since vertical adjustments are restricted. However, as the altitude range increases, 3D-Covert achieves noticeable gains (approximately 15–25%) over 2D-Covert.

This improvement arises because altitude directly affects both path loss and incidence angles in the Lambertian channel model. By increasing altitude near the warden, the UAV reduces h_w and relaxes the covertness constraint, while lowering altitude

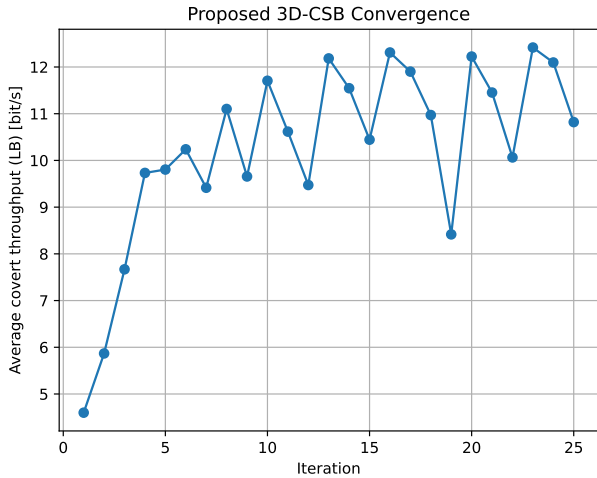


Fig. 4. Convergence behavior of the SCA-based alternating optimization algorithm.

near Bob enhances h_b . This dual effect cannot be achieved with purely horizontal motion, highlighting altitude as a critical degree of freedom in covert VLC.

D. Convergence Behavior

Fig. 4 shows the convergence of the proposed SCA-based algorithm. The average covert throughput increases monotonically and converges within approximately 20–30 iterations. The diminishing improvement across iterations confirms the stability of the convex approximations, consistent with SCA-based UAV optimization methods.

E. Optimized Trajectories

Fig. 5 illustrates the optimized UAV trajectories.

The 3D trajectory leverages altitude adaptively, increasing height near the warden to reduce detectability and decreasing height when approaching Bob to strengthen the communication link. In contrast, 2D-Covert relies solely on horizontal detours, resulting in less effective separation between h_b and h_w .

These results demonstrate that 3D mobility enables more effective geometry shaping, which is essential for improving the throughput–covertiness tradeoff.

VII. CONCLUSION

This paper presents a covert UAV-assisted VLC framework that jointly optimizes the 3D trajectory and transmission intensity under a KL-divergence-based covertness constraint. The optimization problem is formulated by jointly considering the UAV trajectory, the optical transmission intensity, and the covertness requirements. Due to the NP-hard nature of the problem, it is decomposed into two tractable subproblems, and a 3D-CSB algorithm is proposed to solve the studied problem. Numerical results demonstrate that the proposed 3D-CSB algorithm significantly improves covert throughput compared to baseline schemes, highlighting the critical role of altitude control and

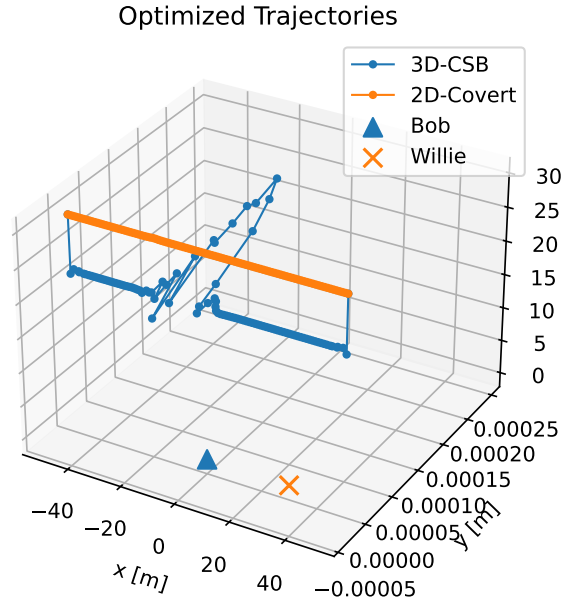


Fig. 5. Example optimized 3D and 2D trajectories.

trajectory design in balancing communication performance and covertness.

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